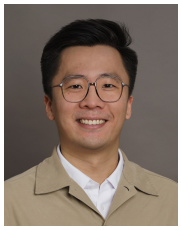
Yifeng Shen



Software Engineer
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Berlin, Germany

EXPERIENCE

EDAG Engineering Group AG Mar. 2022 - Feb. 2026
Software Engineer → Senior Software Engineer (Promoted Dec. 2023) Berlin, Germany

- **Autonomous driving - behavior & trajectory planning, EDAG CityBot** *(4-person ADCU team incl. 1 tech lead)*
  - Developed ROS2 C++ trajectory planning module using MPC (CasADi) for real-time, obstacle-aware navigation
  - Implemented rule-based behavior planning using a finite state machine (FSM) architecture
  - Designed the optimization model, including vehicle and dynamic obstacle modeling with adaptive constraints
  - Integrated multi-threaded ROS2 planning node with vehicle state, map, teleoperation, and vehicle control interfaces
  - Built MATLAB-based simulation with mock map data to validate and visualize planning approaches
  - Supported deployment and integration of planning software on NVIDIA Jetson AGX Xavier-based ADCU platform
- **Linux-based automotive OS validation, Volkswagen VW.OS** *(10-member agile validation team)*
  - Designed and maintained requirement-based test suites across multiple product clusters
  - Developed automated test scripts using Robot Framework and integrated them into Jenkins CI pipelines
  - Created and tracked defects by analyzing logs, supporting CI and daily build validation
  - Implemented shell scripts to improve testing efficiency and automation
- **Embedded system integration & HiL test automation, OTIS** *(Sole full-time engineer working with 1 tech lead)*
  - Delivered the first successful demonstration of the automated HiL pipeline, earning strong approval from stakeholders
  - Developed MATLAB–CANoe TCP/IP interface enabling real-time bidirectional communication
  - Implemented CAPL-based APIs in CANoe for executing control actions and retrieving system states
  - Built Python application for parsing serial data from control board and integrated data into CANoe via pywin32
  - Automated Simulink test model generation from Excel and created Python scripts for execution and reporting

ContiTech, Continental AG Apr. 2021 - Sept. 2021
Intern Hamburg, Germany

- Designed UI/UX prototype in Figma for an RFID-based automotive connector system visualization

E.ON Energy Research Center May 2020 - Jan. 2021
Master thesis Aachen, Germany

- Designed and implemented IEC 61850 to XMPP communication mapping for Smart Grid applications
- Extended a gateway application with XMPP sender/receiver architecture using open-source C library (libiec61850)

NXP Cup National University Students Intelligent Car Race June 2017 - Aug. 2017
Competition Shanghai, China

- Developed vision-based navigation system on ARM Cortex-M4 (Kinetis K60) in C

EDUCATION

Rheinisch-Westfälische Technische Hochschule Aachen Oct. 2018 - Feb. 2022
Master of Science in *Electrical Engineering, Information Technology and Computer Engineering* Aachen, Germany

- GPA: 1.8 (best grade: 1.0)

East China University of Science and Technology Sept. 2014 - July 2018
Bachelor of Engineering in *Automation: Measurement and Control Technology & Instrumentation* Shanghai, China

- GPA: 3.75 (best grade: 4.0), Honor: National Scholarship (Awarded to top 0.2% of students in China)

SKILLS

Languages: Mandarin (Native), English (CEFR C1), German (CEFR B2)
Programming Languages: C, C++, Python, Assembly (Basic), TypeScript (Basic), PLC/Ladder Diagram (Basic)
Robotics, Embedded & Automotive: ROS2, MPC, AUTOSAR, CANoe, IEC 61850, XMPP, Real-time Systems, PLC
Systems & Tools: Git, Linux, CMake, Docker, Jenkins, JIRA, Confluence, VS Code
Testing & Validation: Robot Framework, CodeBeamer, MATLAB, Simulink, Foxglove
Web Development: Next.js, React, Tailwind CSS, Vercel, HTML5, CSS3